

MESKAYE HIZUNAN MEDHANE ALEM MONASTERY SCHOOL
2018 E.C 2nd Semester Chemistry Model Exam For Grade 12th sts.

Time allowed:- 2: 00hr

Name _____ Section _____ Roll No. _____

I. Direction:- This exam paper consist 70 multiple choice questions. Attempt to read all questions carefully then select the correct answer among the given alternative and put the Choice of your answer on the separate answer sheet provided (1Pt each).

- The combined gas law equation can be derived from:-
 - Avogadro's and Graham's laws.
 - Charles' and Graham's laws.
 - Boyle's and Charles' laws.
 - Boyle's and Avogadro's laws.
- Which of the following statements is true about an ideal gas? It is a hypothetical gas that obeys the gas laws
 - at low temperature and low pressure
 - at low temperature and high pressure
 - closely at high temperatures and low pressures
 - closely at high temperatures and high pressures
- The migration or intermingling of molecules of different gases as a result of random molecular motion is known as -----.
 - convection
 - diffusion
 - evaporation
 - osmosis
- If an unknown gas diffuses 1.75 times faster than sulfur dioxide gas, what is the molar mass of the unknown gas? (molar mass of "SO₂" is 64)
 - 112
 - 36.57
 - 48
 - 96
- Non-volatile liquids have little tendencies to evaporate at a given temperature. This is because of they have
 - Relatively large surface area
 - Weaker force of attraction
 - stronger force of attraction
 - relatively lower temperature
- Two gases A and B were applied to the two ends of 40 cm long glass tube. the two gases met at a distance of 30 cm from the side where gas B is applied which statement is true about these gases
 - The rate of diffusion of A is bigger than the rate of diffusion of B.
 - The density of A is smaller than the density of B
 - The rate effusion of B is smaller than the rate effusion of A
 - The malar mass of A is bigger than the molar mass of B.
- Consider the following table that shows different liquids with their boiling points.

Liquid	Boiling point)
Water	100
Ethanol	78
Diethyl ether	35
Toluene	120

Which liquid has the highest vapor pressure?

- Diethyl ether
 - Ethanol
 - Toluene
 - Water
8. The following figure shows the change in temperature as a solid substance is heated.



Which path represents the co-existence of liquid and gas phases?

- a. 3 b. 1 c. 4 d. 2

9. Which of the following combinations correctly matches?

- a. Molar heat of solidification → the amount of heat needed to convert one mole of a solid to a liquid at its melting point.
b. Sublimation → change from a solid to a liquid at a definite temperature.
c. Solidification → change from a liquid to a solid at the freezing point of the liquid
d. Molar heat of fusion → the amount of heat released when one mole of a liquid is converted to a solid at its freezing point

10. Which of the following factors affect the pressure of an enclosed gas?

- a. temperature b. volume c. number of particles d. all of the above

11. Which law states that “volume of a gas is directly proportional to its temperature in kelvins if the pressure and the number of particles is constant.” ?

- a. Boyle’s law. b. Gay-lussac’s Law c. Avogadro’s law. d. Charles’s law.

12. If the volume of a cylinder is reduced from 4.0 L to 2.0 L, the pressure of the gas in the cylinder will change from 100 kilopascals to

- a. 50 kpa. b. 150 kpa. c. 200 kpa. d. 400 kpa.

13. What type of change occurs when water changes from a solid to a liquid?

- a. a phase change c. an irreversible change
b. a physical change d. both A and B

14. During a phase change, the temperature of a substance

- a. increases. c. stays the same.
b. decreases. d. either increases or decreases

15. During what phase change does the arrangement of water molecules become more orderly?

- a. melting b. freezing c. boiling d. condensing

16. Which of the following phase changes is an endothermic change?

- a. condensation b. vaporization c. deposition d. freezing

17. All of the following are phases (states) of matter EXCEPT:

- a. solid b. liquid c. gas d. Vaporization

18. Water is different from other substances because:

- a. It is more dense as a solid than a liquid c. it is more dense as a gas than a liquid
b. It is less dense as a solid than liquid d. It is less dense as a solid than a gas

19. This matter has a fixed shape and volume with particles closely packed together with little movement. It is a:

- a. liquid b. solid c. gas d. plasma

20. Matter changing from a solid to a liquid is called:

- a. evaporation b. sublimation c. deposition d. melting

21. Which of the following is NOT a way that matter changes phase?

- a. melting b. freezing c. evaporation d. mixing

22. Matter changing from a solid to a gas is called:

- a. evaporation b. sublimation c. deposition d. melting

23. The melting point of ice is:

- a. 0°C b. 100°C c. 32°C d. 60°C

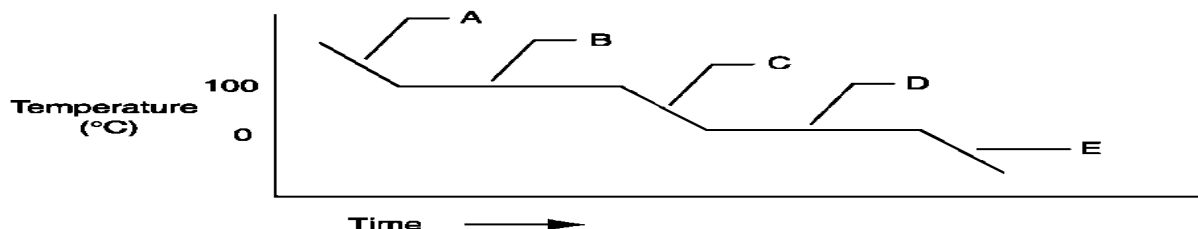
24. In a solid, the particles

- a. overcome the strong attraction between them c. slide past one another
b. vibrate in place d. move independently of one another

25. A gas

- a. has a definite volume but no definite shape
b. has a definite shape but no definite volume c. has fast moving particles
d. has particles that are always close together

Cooling Curve of Water

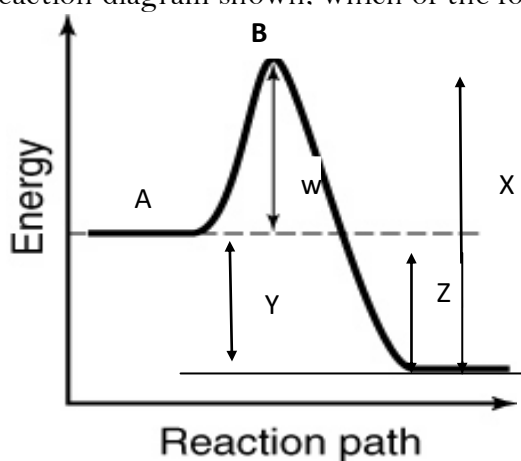


26. Letter D above represents
 - a. the boiling point/freezing point
 - b. the melting point/freezing point
 - c. the boiling point/melting point
 - d. the condensation point/boiling point
27. At higher temperatures
 - a. particles in an object move faster.
 - b. gas particles bump into walls less often.
 - c. a gas contracts.
 - d. particles in an object have less energy.
28. Which of the following occurs when a liquid becomes a gas?
 - a. The particles give off energy.
 - b. The particles break away from one another.
 - c. The particles move closer together.
 - d. The particles slow down.
29. If you open a bottle of perfume, after a period of time, the people on the opposite side of the room will be able to smell it due to the process of
 - a. condensation.
 - b. evaporation.
 - c. sublimation.
 - d. vapor pressure.
30. The melting point of salt is the same as its
 - a. boiling point.
 - b. condensation point.
 - c. freezing point
 - d. sublimation point.
31. A graph that shows the change in temperature of a substance as it is heated will show
 - a. a straight line as the substance melts.
 - b. a straight line as the substance freezes.
 - c. a rising line as the substance melts.
 - d. a falling line as the substance melts.
32. The reverse of condensation is
 - a. boiling.
 - b. freezing.
 - c. evaporation.
 - d. sublimation.
33. When two or more elements join together chemically,
 - a. a compound is formed.
 - b. a mixture is formed.
 - c. a substance that is the same as the elements is formed.
 - d. the physical properties of the substances remain the same.
34. How is a compound different from a mixture?
 - a. Compounds have two or more components.
 - b. Each substance in a compound loses its characteristic properties.
 - c. Compounds are commonly found in nature.
 - d. Solids, liquids, and gases can form compounds.
35. Temperature at which a solid begins to liquefy
 - a. melting point
 - b. boiling point
 - c. heat of vaporization
 - d. freezing point
36. State of matter in which particles are tightly packed together.
 - a. solid
 - b. liquid
 - c. gas
 - d. plasma
37. A solid is a state of matter that has
 - a. a definite volume and indefinite shape.
 - b. a definite volume and definite shape.
 - c. an indefinite volume and indefinite shape.
 - d. an indefinite volume and definite shape.
38. A sample of oxygen occupies 47.2 liters under a pressure of 1240 torr at 25°C. What volume would it occupy at 25°C if the pressure were decreased to 730 torr?
 - a. 27.8 L
 - b. 29.3 L
 - c. 32.3 L
 - d. 47.8 L
 - e. 80.2 L
39. A sample of nitrogen occupies 5.50 liters under a pressure of 900 torr at 25°C. At what temperature will it occupy 10.0 liters at the same pressure?
 - a. 32°C
 - b. -109°C
 - c. 154°C
 - d. 269°C
 - e. 370°C
40. Under conditions of fixed temperature and amount of gas, Boyle's law requires that
 - I. $P_1V_1 = P_2V_2$
 - II. $PV = \text{constant}$
 - III. $P_1/P_2 = V_2/V_1$
 - a. I only
 - b. II only
 - c. III only
 - d. I, II, and III

41. The volume of a sample of nitrogen is 6.00 liters at 35°C and 740 torr. What volume will it occupy at STP?(Take and ideal gas Constant $R=0.0821 \frac{L \cdot atm}{Mol \cdot K}$)
 a. 6.59 L b. 5.46 L c. 6.95 L d. 5.67 L e. 5.18 L
42. ____ is the quantity of heat required to convert one mole of a solid to a gas at its sublimation point.
 a. Molar heat of sublimation c. heat of solidification
 b. molar heat of vaporization d. molar heat of fusion
43. ____ states that the volume of a fixed mass of gas at constant pressure varies directly with the Kelvin temperature.
 a. Boyles Law b. Lussac's law c. Avogadro's Law d. Charles Law
44. The melting point of a solid is the same as _____.
 a. Boiling point b. freezing point c. Heat of fusion d. Heat of sublimation
45. For the reaction $N_2O_4 \rightleftharpoons 2NO_2$; $\Delta H = +14kJ$. The extent of the reaction can be increased by:-
 a. Increasing the pressure c. Increasing the temperature
 b. Using a catalyst d. Introduce inert gas at constant volume
46. In which of the following cases the reaction goes farthest to completion?
 a. $K_c = 10^{-2}$ b. $K_c = 10^3$ c. $K_c = 10^2$ d. $K_c = 10^{-3}$
47. Which one determines the rate of a reaction?
 a. Size of the reactant c. Temperature
 b. Nature of reactants d. All
48. For zero order reaction, the rate of reaction is independent of:-
 a. Surface area c. Nature of reactant
 b. Reactant's concentration d. Temperature
49. For the reaction $A + 3B \rightarrow 2C$, how does the rate of disappearance of B compare to the rate of production of C?
 a. the rate of disappearance of B is 1/2 the rate of appearance of C
 b. the rate of disappearance of B is 3/2 the rate of appearance of C
 c. the rate of disappearance of B is 2/3 the rate of appearance of C
 d. the rate of disappearance of B is 1/3 the rate of appearance of C
50. For the reaction $2A + 3B \rightarrow 4C + 5D$, the rate of the reaction in terms of ΔA would be written as:
 a. $-\Delta A/\Delta t$. b. $+1/2 \Delta A/\Delta t$. c. $-1/2 \Delta A/\Delta t$. d. $-2 \Delta A/\Delta t$. e. $+\Delta A/\Delta t$.
51. In the combustion of methane, $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$, which reactant has the greatest rate of disappearance?
 a. CH_4 b. O_2 c. CO_2 d. H_2O e. CH_4 and O_2 have the same rate of disappearance.
52. A scientist conducts an experiment to determine the rate of the following reaction:
 $N_2(g) + O_2(g) \rightarrow 2NO(g)$ If the initial concentration of N_2 was 0.50 M and the concentration of N_2 was 0.450 M after 0.10 s, what is the rate of the reaction?
 a. 0.500 M/s b. 10.0 M/s c. 1.00 M/s d. 0.250 M/s e. 5.00 M/s
53. A scientist conducts an experiment to determine the rate of NO formation in the reaction:
 $N_2(g) + O_2(g) \rightarrow 2NO(g)$ If the initial concentration of N_2 was 0.50 M and the concentration of N_2 was 0.450 M after 0.10 s, what is the rate of NO formation?
 a. 0.500 M/s b. 10.0 M/s c. 1.00 M/s d. 0.250 M/s e. 5.00 M/s
54. If the rate of appearance of O_2 in the reaction: $2O_3(g) \rightarrow 3O_2(g)$ is 0.250 M/s over the first 5.50 s, how much oxygen will form during this time?
 a. 1.38 M b. 0.25 M c. 4.13 M d. 0.46 M e. 0.69 M
55. Among the following gases, which gas diffuses more rapidly?
 a. Kr b. O_2 c. N_2 d. H_2

56. What is the molecular mass of a gaseous compound having a density of 4.95 g/L at -35°C and 1020 torr?
 a. 85 g/mol b. 120 g/mol c. 72 g/mol d. 24 g/mol
57. A balloon was filled with nitrogen gas which has a volume of 1.5L. How many grams of nitrogen gas would be required to inflate this balloon to a pressure of 1560 torr at 45°C ?
 a. 0.12 b. 11.5 c. 26 d. 3.2
58. Under what conditions do real gases obey gas laws?
 a. High pressure and high temperature c. Low pressure and low temperature
 b. High temperature and low pressure d. High pressure and low temperature
59. Which of the following is correct according to Graham's law of diffusion?
 a. The density of a gas is directly proportional to the square root of the rate of diffusion
 b. The Rate of diffusion of a gas is directly proportional to the square root of its density
 c. The density of a gas is inversely proportional to the square root of the rate of diffusion
 d. The Rate of diffusion of a gas is inversely proportional to the square root of its density
60. "Equal volumes of different gases at the same temperature and pressure contain equal numbers of molecules"; this is a statement of:
 a. Gay Lussac's law c. Avogadro's law
 b. Thomas Graham's law d. Ideal gas law
61. The rate diffusion of CO is greater than that of CO_2 . This is because:
 a. CO_2 is more stable than CO c. The molar mass of CO is less than that of CO_2
 b. CO is more reactive than CO_2 d. The density of CO is greater than that of CO_2
62. Which one of the following conditions can affect the rate of diffusion of gases?
 a. Change in temperature c. Change in the duration or time of diffusion
 b. Change in pressure d. All can affect the rate of diffusion
63. Which of the following is correct about the physical states of matter?
 a. Gases particles are more compacted than liquids and solids
 b. Solid particles vibrate relative to one another but continuously change their position
 c. At a given temperature the kinetic energy of gases is greater than that of liquids and solids
 d. At a given temperature, solid particles possess large volume of space than liquids and gases
64. According to Gay Lussac's law:
 a. The volume of a gas is directly proportional to the kelvin temperature under constant pressure
 b. The pressure and the temperature of a fixed volume of gas are directly related to each other
 c. The volume of a fixed mass of gas is directly proportional to its number of moles
 d. The pressure and the temperature of a fixed volume of gas are inversely related to each other
65. Which one of the following happens when the Kelvin temperature of a fixed amount of gas is increased?
 a. The number of moles of the gas increases c. The velocity of the gas particles increases
 b. The stability of the gas molecules increases d. The collision between the molecules decreases
66. At 27°C and a pressure of 3 atm, a quantity of gas has a volume of 30 L. What will be the volume of the gas under STP conditions?
 a. 60 L b. 40.95 L c. 81.9 L d. 15 L
67. A sample of argon gas occupies 100 L under a pressure of 2280 mm Hg at 30°C . What volume would it occupy at 30°C if the pressure were changed to 1.5atm?
 a. 200 L b. 100 L c. 300 L d. 150 L
68. Particles of matter that collide with the wall of the container with very high rate are that of:
 a. Solids b. Gases c. liquids d. Plasma
69. The energy needed for a reaction to proceed from reactants to products is called _____
 a. collision energy. d. potential energy.
 b. kinetic energy. e. thermodynamic energy.
 c. activation energy.

70. For the reaction diagram shown, which of the following statements is true?



- a. Line W represents the ΔH for the forward reaction; point B represents the transition state
- b. Line W represents the activation energy for the forward reaction; point B represents the transition state
- c. Line Y represents the activation energy for the forward reaction; point C represents the transition state
- d. Line X represents the ΔH for the forward reaction; point B represents the transition state